

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTEEOE505B - ELECTRICAL SAFETY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

- 1.Each question carries 12 marks.
 - 2.Question No. 1 will be compulsory and include objective-type questions.
 - 3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 - 4.Use of non-programmable scientific calculators is allowed.
 - 5.Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which of the following body is responsible for developing Indian standards for electrical safety?
 - 1) Bureau of Indian standards
 - 2) Central electricity commission
 - 3) Central Electricity Authority
 - 4) None of the above
 - 2 Which part of National Electric Code of India explain about protective measures from electric shock
 - 1) Part 3
 - 2) Part 1
 - 3) Part 2
 - 4) Part 4
 - 3 An Indian Standard called _____ lays forth the specifications for heat-resistant work clothes for industrial personnel
 1. IS 15748:2022
 - 2.IS 4770 (1991)
 3. IS 15652:2006
 4. none of the above
 - 4 Which of the following statement is correct?
 1. Proximity testers do not require actual metal to-metal
 2. Contact Testers are testers that make actual metal-to-metal contact with the circuit being energized
 3. Proximity testers do not indicate the actual level of the voltage that is present
 4. all of the above

- 5 What is blast?
 1. the vaporization of solid metal conductors into a gas is an exothermic or heat-releasing event that leads to rapid superheating of the surrounding air
 2. physical stimulation that occurs when electric current flows through the human body
 3. Both a and b
 4. None of the above
- 6 Which of the following is not a standard method of bonding
 1. exothermic welding
 2. Soldering
 3. listed irreversible compression-type connections
 4. listed pressure connections
- 7 Which of the following is/are outcome of reliability centered maintenance analysis:
 1. Redesign
 2. perform condition based monitoring
 3. Perform no maintenance
 4. all of the above
- 8 Which type of maintenance consists of regularly scheduled inspections, adjustments, cleanings, lubrication, and replacement of components and equipment.
 1. Condition based monitoring
 2. real time monitoring
 3. Preventive maintenance
 4. None
- 9 Which of the following Indian standard is designed for rubber gloves
 1. IS 4770 (1991)
 2. IS 15652:2006
 3. IS 732
 4. None of the above
- 10 Which of the following safety team is location specific in a company:
 1. Employee Electrical Safety Team
 2. Company Electrical safety team
 3. Both
 4. None
- 11 Abbreviation of ATPV
 1. Artificial thermal performance value
 2. Arc thermal proactive value
 3. Arc Thermal protection value
 4. Arc Thermal Performance Value
- 12 Which of the following is not a safety device
 1. hot stick
 2. insulated hand tools
 3. rubber gloves
 4. all of the above

Q2. Solve the following.

- A) Explain different parts of National Electrical Code of India which was published in 2023 by Bureau of Indian Standards 6
- B) Discuss different Indian electricity acts related to electrical safety? 6

- Q3. Solve the following.
- A) Explain two use cases of real time monitoring for a processing industries? 6
 - B) List out different principles of reliability centered maintenance? 6
- Q4. Solve Any Two of the following.
- A) Explain any two of the following with application? 6
 - 1) Barriers and signs
 - 2) lockout devices
 - 3) Safety tags
 - B) How to protect working personnel from flash and thermal hazards? Why inspection is important before wearing any protective suit? Also, discuss the role of arc thermal performance values in choosing the protective wearables? 6
 - C) Write a short note on electrician's safety kit? 6
- Q5. Solve Any Two of the following.
- A) Explain the six step safety methods in detail? 6
 - B) Explain hot work decision tree with neat and clean flow chart ? 6
 - C) Explain the purpose of grounding ? Also, discuss different grounding methods for low voltage and high voltage systems ? 6
- Q6. Solve Any Two of the following.
- A) What is safety audit? Explain different steps involved to be performed during safety audit? 6
 - B) Write down procedure of first aid for a unconscious electric shock victim with symptoms 6
 - C) what are the different rescue techniques? Write down the general rescue procedure? 6

*** End ***